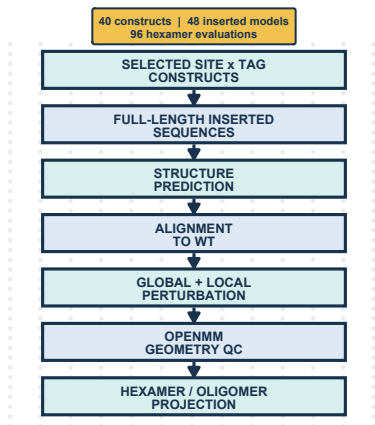


a

Explicit insertion-modeling workflow

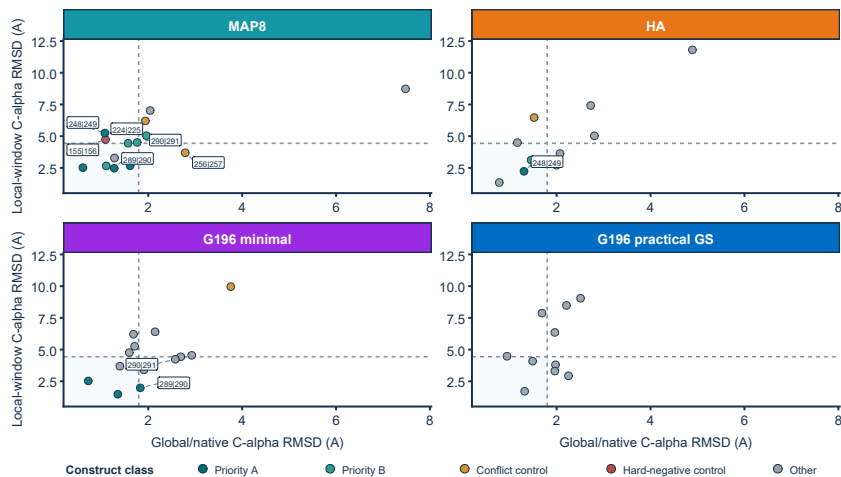
48/48 tagged models completed geometry QC



b

Global structural perturbation landscape

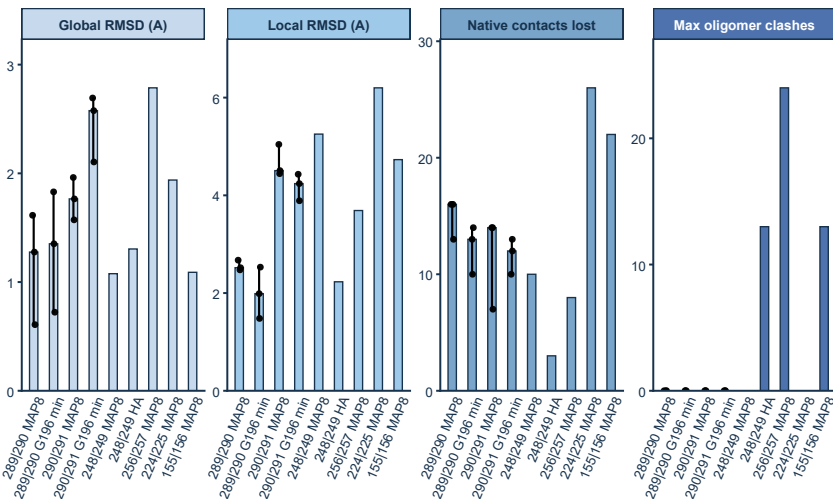
Tag identity is encoded by panel colour; small circles are individual models. Point fill denotes construct class; dashed lines show global medians.



c

Focal construct comparison

Bars: model median; n=1 bars are single-model values. Black dots + min-max whiskers: n=3 computational predictions only.



d

Representative WT overlays

WT light grey; inserted chain blue; tag orange.

289/290 x MAP8 | lower local perturbation



290/291 x MAP8 | adjacent-junction contrast



248/249 x MAP8 | global/local discordance

